RELATIVELY VERY AMPLE IMPLIES RELATIVELY AMPLE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file morphisms.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The chapter file pinned-original-source.tex carries only \input{preamble} on its first line, so no preamble text lives inside the chapter. The retained file sources/stacks-morphisms/retained-preamble.tex is a mechanical, line-for-line copy of the project's single shared preamble as staged in the snapshot. Nothing was added, removed or redefined, and the excerpt serves only as preamble, never as a span coordinate.

Editorial note. Both counted spans start at the \begin line of their environment, so the statement and the author's proof are delivered whole; surrounding prose of the section is deliberately outside the counted spans and is carried as standard prerequisites instead.

Editorial note. The citation fields (source_id, source_version, authors, source_url, license.name, source_title) are identical to those of every other card this lane has produced from this chapter, so the batch selector treats them as one source.

Editorial note. Read in full under the Lead's 12:4x rule (screening hit re-read, pb 1060B). The author writes verbatim 'We omit the verification that the morphism r_{L,psi} associated to this map is exactly the immersion i.' - a routine identification of the constructed morphism with the given immersion; everything else in the proof (the construction of A, psi and the appeal to part (6) of the characterisation lemma) is carried out, so this is not a hint-only proof.

Allowed standard prerequisites (editorial).

- Part (6) of the characterisation of relative ampleness. Lemma \ref{lemma-characterize-relatively-ample}, cited at proof lines 9033-9034. Gives relative ampleness once the associated morphism is an immersion.
- The relative Proj of a symmetric algebra is the projective bundle. Constructions, Section \ref{constructions-section-invertible-relative-proj}, used at proof lines 9020-9022. Identifies the target of the immersion.

Source reference. Original lemma 01VN: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2452–2452. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 01VJ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2448–2448. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original reference cite-EGA: sources/stacks-morphisms/pinned-original-source.tex, source lines 8997–8997. Bibliographic pointer appearing in the counted text; it carries no mathematical argument.

Date: stacks-project:morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.2. *Let $f : X \to S$ be a morphism of schemes. Let $\mathcal{L}$ be an invertible $\mathcal{O}_X$-module. If f is quasi-compact and $\mathcal{L}$ is a relatively very ample invertible sheaf, then $\mathcal{L}$ is a relatively ample invertible sheaf.*

Proof. By definition there exists quasi-coherent $\mathcal{O}_S$-module $\mathcal{E}$ and an immersion $i : X \to \mathbf{P}(\mathcal{E})$ over S such that $\mathcal{L} \cong i^*\mathcal{O}_{\mathbf{P}(\mathcal{E})}(1)$. Set $\mathcal{A} = \mathrm{Sym}(\mathcal{E})$, so $\mathbf{P}(\mathcal{E}) = \underline{\mathrm{Proj}}_S(\mathcal{A})$ by definition. The graded $\mathcal{O}_S$-algebra $\mathcal{A}$ comes equipped with a map

$$\psi : \mathcal{A} \to \bigoplus\nolimits_{n\geq 0} \pi_*\mathcal{O}_{\mathbf{P}(\mathcal{E})}(n) \to \bigoplus\nolimits_{n\geq 0} f_*\mathcal{L}^{\otimes n}$$

where the second arrow uses the identification $\mathcal{L} \cong i^*\mathcal{O}_{\mathbf{P}(\mathcal{E})}(1)$. By adjointness of f_* and f^* we get a morphism $\psi : f^*\mathcal{A} \to \bigoplus_{n\geq 0} \mathcal{L}^{\otimes n}$. We omit the verification that the morphism $r_{\mathcal{L},\psi}$ associated to this map is exactly the immersion i. Hence the result follows from part (6) of Lemma 01VJ. □